

Area Courses

Controls

Fundamental Courses

EE 482 Linear Control Systems -or-
AME 451 Linear Control Systems I
 4 units

EE 543 Digital Control Systems

4 units

EE 482

EE 585 Linear System Theory

4 units

EE 441 or EE 510

Mathematical Foundations

EE 503 Probability for Electrical and
 Computer Engineers
 4 units

EE 510 Linear Algebra for Engineering

4 units

EE 562 Random Processes in
 Engineering

4 units

EE 503, EE 510

EE 588 Optimization for the Information
 and Data Sciences

4 units
 EE 503

EE 510

Nonlinear and Robust Control

EE 587 Nonlinear Control Systems

4 units

EE 482, EE 503

EE 585

EE 593 Robust Multivariable Control

4 units

EE 587

EE 482

EE 585

Stochastic, Decentralized, and Network Control

EE 556 Stochastic Systems and
 Reinforcement Learning

4 units

EE 512

EE 503

EE 648 Game Theory with Engineering
 Applications

4 units

EE 503

EE 649 Stochastic Network Optimization &
 Adaptive Learning for Discrete Time Systems

4 units

EE 503

EE 652 Low-Power Wireless Networks

3 units

CSCI 402

EE 450

Cyber-Physical and Complex Systems

EE 520 Introduction to Quantum
 Information Processing

3 units

EE 510

EE 527 Net-Centric Power-System
 Control

4 units

EE 585, 593, 562 or 512

EE 482, EE 510

EE 539 Engineering Quantum
 Mechanics

4 units

EE 554 Cyber-Physical Systems: A
 Computing Perspective

4 units

Financial Engineering

EE 512 Stochastic Processes for
 Financial Engineering

4 units

EE 503, EE 510

EE 518 Mathematics and Tools for
 Financial Engineering *

4 units

ISE 563 Financial Engineering *

3 units

* Available to MSFINE majors only

Legend

Grouping

EE 000 Course Title

Course Units

Recommended Prep.

Prerequisite Courses

Corequisite Courses

This chart shows course relationships

Please check the University Catalogue for specific course details including any recommended preparatory courses and Degree Requirements